TOMMASO CRIPPA

High Performance Computing Engineering Student

crippa.tommaso@gmail.com | Crippius | tommaso-crippa

EDUCATION

Double Master's Degree in High Performance Computing — **Politecnico di Milano - University of Luxembourg**
29.25 / 30 — 2024 - present

- Selected among top candidates for the exclusive EUMaster4HPC double degree program

Bachelor degree in Computer Engineering — **Politecnico di Milano**
110 cum laude / 110 — 2021 - 2024

- Won "Migliori Matricole" scholarship for being one of Politecnico di Milano's best freshmen students of the 21/22 season.

Relevant courses completed:

- **Time Series Analysis**
- **Parallel Computing**
- **Artificial Neural Network and Deep Learning**
- **Streaming Data Analytics**
- **Natural Language Processing**
- **Software Engineering Environments**

EXPERIENCES

Master's Thesis Student @ MEGWARE Computer Vertrieb und Service GmbH — **Chemnitz**, Germany
HPC | **Machine Learning** | Time-Series | SLURM | Linux — February 2026 - present

- Developing a performance analysis tool to automatically detect bottlenecks in HPC applications and providing optimization recommendations for HPC users.
- Processing high-resolution multivariate time series metrics with heuristics and ML approaches to detect anomalies and validating them on real-world HPC workloads.

HPC Ambassador @ EUROHPC Summit — **Krakow**, Poland
HPC | **AI** | **Fluid Dynamics** — March 2025

- Represented the EUMaster4HPC program and the Leonardo and Discoverer supercomputers by doing live demonstrations to the general public of fluid dynamics simulations and Large Language Models deployment.

Operations & Logistic @ Google Developer Group On Campus - Polimi — **Milan**, Italy
AI | **Cloud** | **Mobile** — August 2024 - August 2025

- Actively contributed to the planning and execution of various technical events, including organizing talks, workshops, and hackathons focused on emerging technologies such as Artificial Intelligence Cloud Computing and Mobile Development.

PROJECTS

Early Alzheimer Detection with Deep Learning
Python | pytorch | keras | nilearn | **C++** | libtorch | clinica | SLURM — justvicthor/Early_Alzheimer_Detection_DL

- Developed a deep learning model for early detection of Alzheimer's disease, implementing data preprocessing and augmentation pipelines to analyze medical imaging data with gradCAM visualizations and improve diagnostic accuracy on MeluXina supercomputer.

Whatsapp Wrapped
Python | matplotlib | pandas | fpdf | Flask | HTML | CSS | **Javascript** | Vercel | Render — Crippius/whatsapp-wrapped

- Developed a script that, given an extracted Whatsapp chat, returns an informative visualization of the data hidden inside their conversations, and deployed it on a fullstack website to make this tool available to the general public.

High-Performance Graph Coloring with Branch-and-Bound Framework
Python | MPI | SLURM — Rudolfovoorg/EuroHPC_Student_Challenge_2025_Team_3

- Won a competition where we were tasked to design and implement an optimized Branch-and-Bound algorithm to solve the graph coloring problem, integrating heuristics and parallelization techniques to improve scalability and minimize execution time on VEGA cluster.

SKILLS AND LANGUAGES

Programming Languages:

- **Python** (Advanced)
- **C** (Advanced)
- **C++** (Academic)
- **Java** (Academic)
- **Javascript** (Academic)

Skills:

- **Git** (Advanced)
- **SLURM** (Advanced)
- **DVC** (Advanced)
- **Kafka** (Academic)
- **Spark** (Academic)

Languages:

- **Italian** (Native)
- **English** (Proficient): IELTS 7.5
- **German** (Beginner): A2